

National University of Computer and Emerging Sciences

Software Engineering (CS-3009)/Fundamental of Software Engineering (CS-2004)

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Course Instructor(s)

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Sessional I- Exam

Total Time: 1 Hour

Total Marks: 25

Total Questions: 3

Semester: SP-2025

Campus: Karachi

Dept: Computer Science

Student Name

Roll No

Section

Student Signature

Do not write below this line

Attempt all the questions. Attempt all questions on the answer script according to the sequence, any question answered other than the right sequence will NOT be graded!!

CLO-1: Apply suitable process models and activities for medium size software systems

Question 1:

[1*10=10 Marks]

Answer the following questions in 1 to 2 lines only.

- i) What role do stakeholders play in managing change?

Ans: Stakeholders influence, support, and ensure smooth implementation of organizational change.

- ii) How does the V-Model address testing at different levels of the software development lifecycle?

Ans: The V-Model integrates testing at each development phase for validation.

- iii) What strategies can be employed to manage technical debt (Re-Work) when frequent changes are made to large-scale software systems?

Ans: Refactor regularly, automate testing, modularize code, document changes, enforce standards.

- iv) Differentiate between SCRUM Master and Product Owner.

Answer: A Scrum Master leads the Agile development team and supports the Product Owner by relaying updates to relevant employees. Product Owners manage the product backlog and ensure the company gains maximum value from the product.

- v) In which phase of the requirement engineering, software engineers work with stakeholders to find out about the application domain, work activities, the services and system features that stakeholders want, the required performance of the system, hardware constraints, and so on.

Answer: Requirement Elicitation

CLO-1: Apply suitable process models and activities for medium size software systems

Question 2:

a) A global banking corporation wants to develop a new international banking system to facilitate seamless transactions across borders. The system must integrate with existing banking infrastructure, ensure compliance with international regulations. Multi-Currency Support enable transactions in multiple currencies, including real-time currency conversion. Cross-Border Payments facilitate fast and secure cross-border payments, complying with international regulations. Implement real-time transaction monitoring to detect and prevent fraudulent activities. Ensure the system is available 24/7, with minimal downtime and quick recovery from failures. Integrate the proposed system with existing banking infrastructure, including core

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banking systems and payment gateways and ensure the system provides robust security measures to protect customer data and prevent fraudulent activities and can scale to handle increasing transaction volumes and user growth.

Provide an application for customers to manage their accounts, track transactions, and access customer support and implement advanced security measures, including encryption, firewalls, and intrusion detection systems. Proposed system should ensure compliance with international regulations, including anti-money laundering (AML) and know-your-customer (KYC) regulations. The system's alignment with the required standards will be ascertained through the execution of a thorough quality control and testing methodology, replete with multiple validation checkpoints. Following an exhaustive analysis of stakeholder expectations, market trends, and regulatory requirements, a comprehensive set of system requirements has been distilled without deviation, crystallizing the essential characteristics of the proposed system. Stakeholder Concerns include Meeting regulatory requirements, ensuring security and compliance, Managing complexity, and Meeting customer expectations along alignment with the required standards.

i) Recommend a suitable methodology, either a linear (planned) approach or an iterative one. **[01 Marks]**

ii) Given the complexity and criticality of this project select the most suitable software development process model for this project. **[01 Marks]**

iii) Justify your answer with reference to the case study. **[02 Marks]**

Solution:

1. Planned approach: A planned approach because requirements are already well defined without any deviation and it helps break down complexity, manage stakeholder expectations, ensure compliance with international regulations, identify and mitigate potential risks using testing and verification strategies and provide predictability to ensure timely delivery and adherence to required standards.

2. V Model

3. The V-Model is the most suitable process model for this case study because it provides a structured and disciplined approach to developing a complex system like the international banking system. The V-Model's emphasis on testing and validation at each stage ensures that the system meets the required standards and regulatory requirements, including anti-money laundering (AML) and know-your-customer (KYC) regulations. Additionally, the V-Model's sequential approach ensures that the system is developed in accordance with the specified requirements without deviation, aligning with the stakeholders' expectation of no changes to the initial requirements. This approach also enables the detection and prevention of errors and defects early on, reducing the risk of costly rework and ensuring the delivery of a high-quality system.

b) A university's IT department is developing a mobile app for students to access their grades, attendance, and course materials. The project team initially planned everything upfront, but they are now facing multiple challenges. Professors and students frequently request new features, making the original plan outdated. The team works for months before showing a prototype, and stakeholders often request major changes late in development. Developers, designers, and testers work in silos, leading to miscommunication and delays. The team struggles to decide which features to develop first, leading to wasted effort.

i) As an Agile coach, which Agile Management Framework would you recommend to solve these challenges? **[1.5 Marks]**

ii) Explain how your chosen practice(s) would address the specific problems faced by the team. **[1.5 Marks]**

a. As an Agile coach, which Agile Management Framework would you recommend to solve these challenges?

Solution: Changing Requirements → Use Scrum's Iterative Development

b. Explain how your chosen practice(s) would address the specific problems faced by the team.

Solution:

1. Use Sprint Reviews & Continuous Delivery

2. Team Collaboration

CLO-2: Analyze software requirements and how to produce software design and architecture

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Question 3: Read the scenario carefully and answer the questions provided at the end: **“Baby Daycare Application Requirements”**

Emma is a working mother who recently enrolled her 2-year-old daughter, Lily, in a local daycare. She seeks a reliable mobile application that allows her to monitor Lily’s daily activities, receive real-time updates, and communicate seamlessly with daycare staff. Meanwhile, daycare management requires an efficient system to handle enrollments, track attendance, and manage caregiver schedules. The users are non-tech savvy so the overall burden of requirement understanding relies on the skills of the development side. They deduce the requirements and deliver the specification document to the development team. The application should support multiple user roles, including Parents, Caregivers, and Administrators, with role-based access to ensure appropriate data security. It must comply with child protection regulations like COPPA, GDPR etc. Designed for non-tech- savvy users, the interface must be intuitive, user-friendly, and offer a delightful user experience. The application must be available on both Android and iOS platforms, along with a web version tailored for daycare administrators. Parents and caregivers should be able to create and manage accounts using email, phone numbers, or social media logins. To enhance security, the system must support multi-factor authentication and encrypt all sensitive data, including child profiles and payment details. Parents should have the ability to update their child’s biodata, including food preferences and allergies, track their child’s daily activities, receive notifications for upcoming events and payments, and communicate with caregivers via chat. Additionally, parents should be able to view live streaming of their child’s activities without any performance degradation. The system must be up 99.9% of the time and capable enough of supporting thousands of concurrent users efficiently. Caregivers will have access to a child’s profile to provide personalized care, log daily updates on feeding and health issues, communicate with parents, and send alerts to the daycare’s health services when needed. With this application, Emma can stay informed about Lily’s well-being, receive timely updates, and maintain direct communication with caregivers, giving her peace of mind while at work. Simultaneously, the daycare benefits from a well- organized system that streamlines its operations, improving the overall experience for both staff and parents.

i) List any 5 functional requirements for the user roles specified in the above scenario. **[2.5**

Marks]

ii) Mention all the possible nonfunctional requirements and to which category they fall in, with respect to the given scenario. **[04 Marks]**

iii) Which requirements elicitation technique would be more effective in designing such application and why? Support your answer with strong reasons. **[1.5**

Marks]

Solution:

A1)

Parents:

- Create account
- Login via multiple options (each is a separate requirement)
- Update child profile
- Track daily activities
- receive notifications
- Chatting with caregivers
- view live str eaming

Caregivers:

- Create account
- Login via multiple options (each is a separate requirement)
- View child profile
- Maintain log for all activities
- Chat
- Alert sent for emergency

A2)

- Security (product and regulatory)

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Usability (product)

Portability (product)

Availability (product)

Performance (product)

A3) ethnography would be more effective since the users are no technical so they might not have the complete idea of what the application requirements are.